



Dayananda Sagar University School of Engineering

Devarakaggalahalli, Harohalli Kanakapura Road, Bangalore South Dt., Karnataka 562112

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY

B.Tech Final Year Capstone Project Phase 1-Weekly Progress Report

Project Semester: 7

Aug-Dec 2026

Week No: 4

Date : From : 24/08/2026 To 29/08/2026

Department	Computer Science and Technology
Team Number	Team 3
Names & USN of Team Members	Putta Sujith & ENG23CT0058 K Vikas Aneesh Reddy & ENG23CT0052 Karnati Mokshith & ENG23CT0053
Name of the Guide	Dr. Ramandeep Kaur
Project Title	IntentCloud - Intent-Aware Cognitive Cloud Memory System

Work Carried out by the team during the Week (in Bullet points only):

- Prepared and delivered the Review-1 presentation covering architecture, module design, workflow diagrams, database/index design, technology stack, and task allocation.
- Implemented the POST /search endpoint using Phi-3 Mini to process natural-language queries.
- Developed the Next.js /search page to display the initial search results.
- Added the Tesseract OCR fallback for scanned PDF documents.
- Implemented duplicate detection during file upload.
- Performed regression testing of the upload, extract, embed, and store pipeline.

Individual Contribution from the Student:

Putta Sujith	● Implemented the POST /search endpoint using Ollama and Phi-3 Mini to process natural-language queries. ● Parsed user queries into structured intent information containing topic and keywords. ● Added the Tesseract OCR fallback for scanned or image-only PDF documents. ● Prepared and delivered the Review-1 presentation covering architecture, module design, workflow diagrams, database/index design, technology stack, and task allocation.
K Vikas Aneesh Reddy	● Performed regression testing of the Phase 1–2 upload, extract, embed, and store pipeline. ● Implemented cosine-similarity based duplicate/near-duplicate detection

	during file upload. ● Tested the duplicate detection process to identify highly similar files before redundant indexing.
Karnati Mokshith	● Developed the Next.js /search page with a search bar for natural-language queries. ● Connected the search page with the POST /search API and displayed the initial top-N search results. ● Tested the search interface and verified that the returned results were rendered correctly.
Work Planned for the Following Week: ● Implement sparse/BM25-style keyword scoring alongside dense retrieval. ● Implement Reciprocal Rank Fusion (RRF) to merge dense and sparse rankings. ● Integrate Qdrant hybrid search functionality. ● Add the cross-encoder reranking step and return the final top-3 results. ● Update the Next.js /search page to display reranked top-3 results with highlighted matched snippets. ● Prepare the 30–50 query evaluation set with independently authored ground-truth target files. ● Place the Raspberry Pi hardware order by 5 September 2026.	

Percentage Work Completed: 40%

Signature of the Students:

Signature of the Guide(s)